

Applied Research and Data-Driven Decision.

A/B testing, causal reasoning, and statistical inference in business practice.

Statistical

Analytical

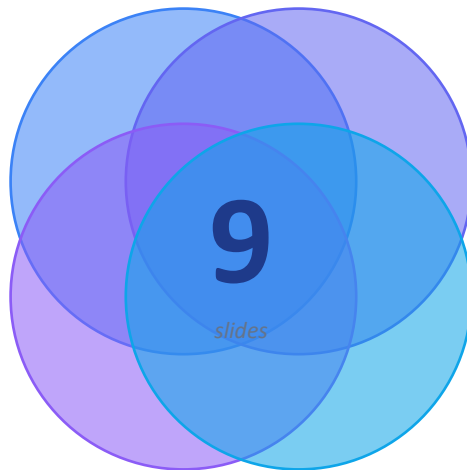
Philosophical

Managerial

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Five philosophical anchors of applied research.

Every decision model rests on assumptions about reality, knowledge, method, values, and duty.



Ontology

What is real

Sales, prices, and preferences exist as measurable events.



Epistemology

How do we know

Truth claims from randomization, replication, and inference, not intuition.



Methodology

How do we study

A/B testing plus regression, complementary tools for causal and predictive claims.



Axiology

What is valuable

Customer welfare, staff workload, margin, and long term brand equity.



Deontology

What is right

Duty to be transparent, no deceptive framing, no manipulation of participants.

The two case studies below anchor each pillar in a concrete Indonesian retail decision.

The data-driven decision cycle.

Six stages from question to learning, each a checkpoint that prevents leaping to conclusions.



1. Question

Frame a decision, name the outcome, define success.

1



2. Design

Choose a study design, randomize where feasible, plan sample size.

2



3. Data

Instrument, collect, and validate. Guard against selection bias.

3



4. Analysis

Descriptive first, then inferential. Report effect size and CI.

4



5. Decision

Translate stats into money and action. Weigh risk, ethics, and cost.

5



6. Learn

Ship, monitor, and iterate. Every rollout is a new experiment.

6

Every decision becomes the next question. The cycle is continuous.

A/B testing in one page, from randomization to decision.

Randomization neutralizes confounders. The test statistic separates signal from noise.

Hypothesis and statistic.

$$H_0 : \mu_T = \mu_C$$

$$H_1 : \mu_T > \mu_C \quad \alpha = 0.05$$

Welch t-statistic

$$t = (\bar{x}_T - \bar{x}_C) / \sqrt{s_C^2/n_C + s_T^2/n_T}$$

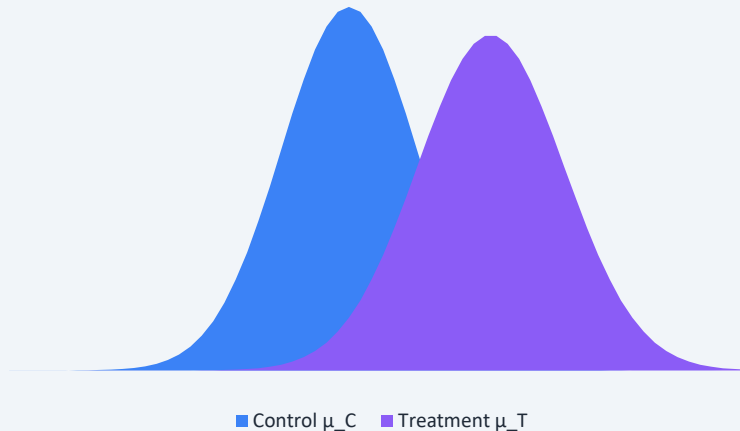
Effect size, Cohen d

$$d = (\bar{x}_T - \bar{x}_C) / s_{pooled}$$

Decision rule

Reject H_0 if $p < \alpha$, act if effect is material.

Signal vs noise, two-sample view.



Overlap = uncertainty. Small p means overlap is unlikely under H_0 .

Case 1, does a 20 percent price promo lift Kopi Susu Aren sales?

Randomized cluster A/B test, 60 outlets in Jabodetabek, 14 days.



Product

Kopi Susu Aren, 22oz



Test units

60 outlets, 30 treatment + 30 control



Randomization

Stratified by region, Jakarta, Bekasi, Depok, Tangerang, Bogor



Window

14 days, Senin to Minggu, two weeks



Primary metric

Cups sold per outlet per day

Price contrast.

Control

Rp 22.000

Regular price

Treatment

Rp 17.600

20 percent off

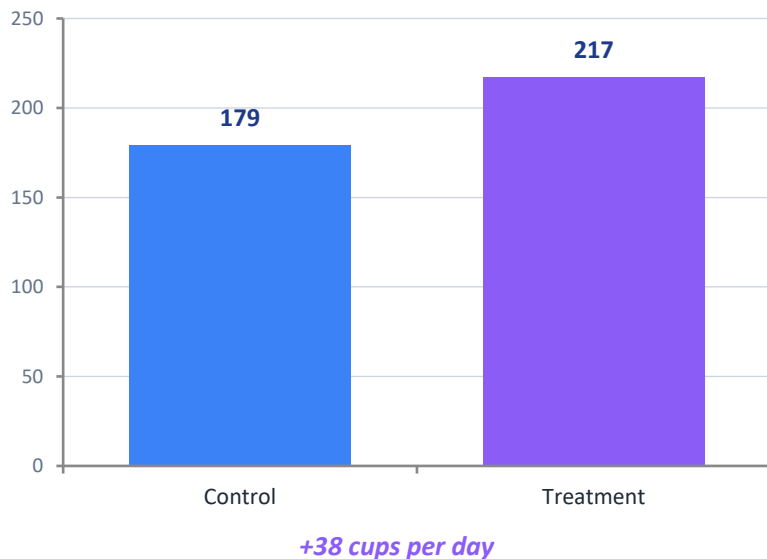
Risk view. A price cut only pays off if volume rises enough to offset revenue per cup.

Sample size, $n = 30$ per arm, ex-ante power 0.80, alpha 0.05, two-sided.

Case 1 result, treatment lifts cups sold by 38 per outlet per day.

Welch $t = 7.36$, $df = 57.9$, $p < 0.001$, Cohen $d = 1.90$. Large effect, high confidence.

Mean cups sold, control vs treatment



Welch two-sample t-test.

Quantity	Symbol	Value
Mean difference	$\bar{x}_T - \bar{x}_C$	38.03 cups
Standard error	SE	5.17
Welch t statistic	t	7.36
Degrees of freedom	df	57.9
Two-sided p value	p	8.1e-10
95 percent CI	CI	[27.7, 48.4]
Effect size	$Cohen\ d$	1.90

Business impact. Revenue delta $\approx$ **Rp 33 juta** per outlet per month at rollout, cost of discount already netted.

See workbook, sheet 03_Case1_Analysis, for live formulas.

Case 2, isolating promo impact when many things move at once.

Indomie Goreng at 30 Indomaret outlets, 60 days, 1,800 outlet-day observations.

Structural model.

units_sold

$$= \alpha + \beta_1 \cdot \text{promo} + \beta_2 \cdot \text{price_rp} \\ + \beta_3 \cdot \text{rainfall} + \beta_4 \cdot \text{payday} \\ + \beta_5 \cdot \text{competitor} + \varepsilon$$

Estimated with OLS via LINEST array formula in Excel.

Identification. Promo timing varies across outlets and days, base price varies by outlet tier, promo depth ranges 10 to 25 percent. This breaks the perfect collinearity between promo and price.



y units_sold

Daily units per outlet



x₁ promo

Binary, 0 or 1, shelf tag active



x₂ price_rp

Shelf price, Rp 2.900 to 3.800



x₃ rainfall_mm

Daily rainfall in mm at outlet



x₄ payday_flag

1 if day 25 to day 3 next month



x₅ competitor_promo

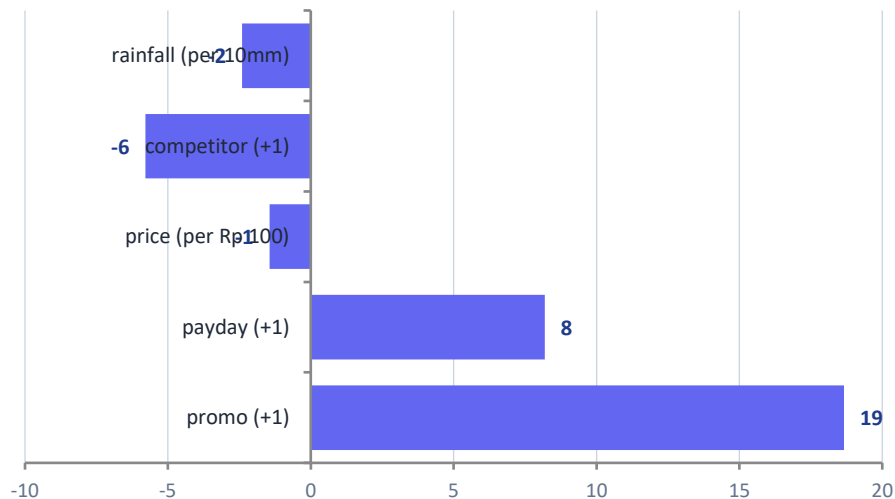
1 if Alfamart within 500m on promo

Six variables, one panel dataset, one causal question, does promo still lift once we control for the rest?

Case 2 result, promo remains a positive driver after controls.

$R^2 = 0.73$, all five predictors significant at $p < 0.001$, price is inelastic at $\epsilon \approx -0.70$.

OLS coefficients, effect on units per outlet per day



Price elasticity at mean.

$$\epsilon = (dQ/dP) \times (P / Q) = -0.70$$

Category is inelastic. Absolute lift comes from promo salience, not price sensitivity.

ROI, national rollout, 20.000 outlets, 30 days.

Extra units per outlet per day	+18.7
Extra gross margin per outlet per day	Rp 44.794
Total incremental margin	Rp 26,88 miliar
Promo execution cost	Rp 240 juta
Net incremental margin	Rp 26,64 miliar

Decision, roll out, monitor by outlet, retest quarterly.

Synthesis, philosophy meets practice.

Where each philosophical anchor lands in the two case studies, and what to do next.

Philosophical audit of the analysis.

Ontology	Sales are countable events tied to price actions.
Epistemology	Truth from randomization and inference, not intuition.
Methodology	A/B test for causal claim, regression for context.
Axiology	Weigh customer welfare, staff workload, and franchisee revenue.
Deontology	Transparent promo framing, no participant manipulation.

Recommendations.

-  **Ship Case 1 promo**
Roll out with weekly monitoring, sunset if lift falls below Cohen $d = 0.5$.
-  **Anchor Case 2 promo on payday windows**
Combine day-25 to day-3 with promo, expected additive lift +26 units per day.
-  **Avoid competitor promo overlaps**
Detect competitor promo signals and pause our promo to defend margin.
-  **Instrument, learn, repeat**
Every rollout is a natural experiment. Log promo, price, weather, and competition.

From evidence to decision, from decision to learning. Terima kasih.